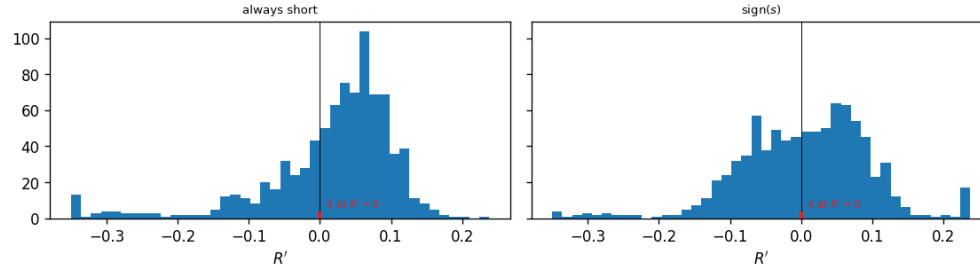


0DTE nearest-OTM package: per-rule return summary, entry 12:30 ET, one-bar hold, models compared on the same 866 days

	n	mean	std	min	max	skew	ex. kurt	t	Sharpe _{clk}	n_{buy}	%buy	Sharpe _{clk} ^x
<i>Short volatility (always short)</i>												
all models (no forecast)	866	0.013	0.115	-1.23	0.23	-3.74	25.06	3.29	1.78	0	0.0	-2.88
sign(s) (± 1 on the sign of s)												
baseline (HAR + calendar OLS)	866	0.007	0.116	-1.23	0.78	-0.64	23.10	1.68	0.91	379	43.8	-3.75
block-diagonal ridge	866	0.005	0.116	-1.23	0.78	-1.21	23.08	1.19	0.64	369	42.6	-4.01
block-diagonal ridge, without the FOMC columns	866	0.005	0.116	-1.23	0.78	-1.18	23.11	1.33	0.72	395	45.6	-3.94
LightGBM [†]	866	0.001	0.116	-1.23	0.78	-1.28	22.87	0.36	0.19	385	44.5	-4.45
XGBoost [†]	866	0.002	0.116	-1.23	0.78	-1.28	22.90	0.45	0.25	391	45.2	-4.40
lasso (causally tuned) [†]	866	0.002	0.116	-1.23	0.68	-1.85	22.93	0.46	0.25	406	46.9	-4.39
lasso ($\alpha = 10^{-4}$)	866	0.004	0.116	-1.23	0.78	-1.20	23.04	1.05	0.57	371	42.8	-4.09
elastic net (causally tuned) [†]	866	0.002	0.116	-1.23	0.68	-1.89	22.92	0.40	0.22	389	44.9	-4.42

block-diagonal ridge, midpoint fills: entry 12:30 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



Distribution of the return of the position taken at 12:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 12:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged. One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean}/\text{std}$, with no autocorrelation correction.

“ex. kurt” is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of that clock’s series: one 30-minute hold per expiration day. It answers “if I only entered at this clock, what is my annual Sharpe?” It is *not* the book’s $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean}/\text{std}) \times \sqrt{252}$ of $R_d^{\text{day}} = \sum_t R'_{d,t}$, the sum of the day’s twelve holds. That reserved name is the book’s daily P&L after pooling the session. At 15:30 the two coincide (one trade that day); they do not on 10:00–15:00 or on the pooled page. Sharpe^x is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the twelve positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook’s column. The other column this table adds to the deck’s is $\text{Sharpe}_{\text{ann}}^x$, the same rule’s annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck’s.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). [†] marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}_{\text{hr}}^2 h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day’s remaining-session share of realized variance, $RV_t / \sum_{s \geq t} RV_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck’s treatment is used here too — the volatility that reproduces the package midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat.